Ufac

AST101

Introductory Astronomy

Unit II

The solar system

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PART 1

The planets

Orbits

Kepler's laws

→ The planets move on ellipses, with the Sun at a focus
→ A planet sweeps equal areas in equal times
→ The square of the period grows with the cube of the orbit

Definition

An astronomical unit (au) is the mean distance from the Earth to the Sun, about 1.5×10^8 km.

Tip. In years and astronomical units, Kepler's third law is just $T^2 = a^3$.

One step at a time

→ The orbit of a planet is an ellipse

→ The Sun sits at one of its foci

Kepler's first law

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Example 1.1 — The period of Mars

Mars orbits the Sun at $a = 1.52$ au. How long is its year?

Solution. From $T^2 = a^3$:

$$T = \sqrt{1.52^3} \approx 1.87 \text{ years}$$

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Exercise 1.1 — Jupiter and Saturn

Find the period of:

1. Jupiter, at 5.2 au
2. Saturn, at 9.5 au
3. an asteroid at 2.8 au

Answer. About 11.9, 29.3 and 4.7 years.

PART 2

On the computer

The periods in Python

Python

```python
planets = {"Mercury": 0.39, "Earth": 1.00, "Mars": 1.52}

for name, a in planets.items():
    print(name, round(a ** 1.5, 2))
```

Output

```
Mercury 0.24
Earth 1.0
Mars 1.87
```

Tip. `a ** 1.5` is the same as `sqrt(a ** 3)`.

Thank you!

Next lecture: the Moon

Read chapter 3 of the book.